CYBERSECURITY HISTORY

NOVEMBER 1988

The Morris Worm

The internet's first major cyber attack — a graduate student's experiment that brought thousands of machines to their knees and changed computer security forever.

Andrea S. Atzeni
< shocked @ polito.it >

The Man Behind the Worm

Robert Tappan Morris

Robert Tappan Morris, a Cornell University graduate student and son of NSA cryptographer Robert Morris, created the worm reportedly just to see if it could be done.

He launched it from **MIT's network** at 8:30 pm on November 2, 1988 — deliberately choosing MIT to avoid the worm being traced back to Cornell.

rumors he worked with a friend or two:

Paul Graham, then a Harvard student, allegedly sent Morris mail asking for "any news on the brilliant project."

Technical Specifications

Language

Written in C

Targets

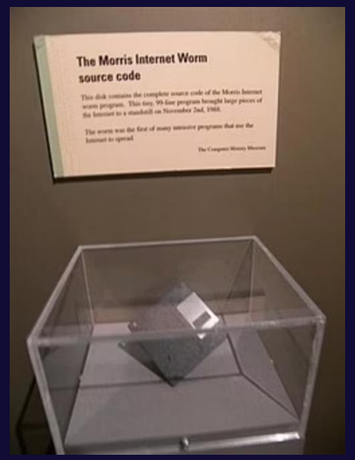
DEC VAX & Sun-3

OS

4 BSD Unix

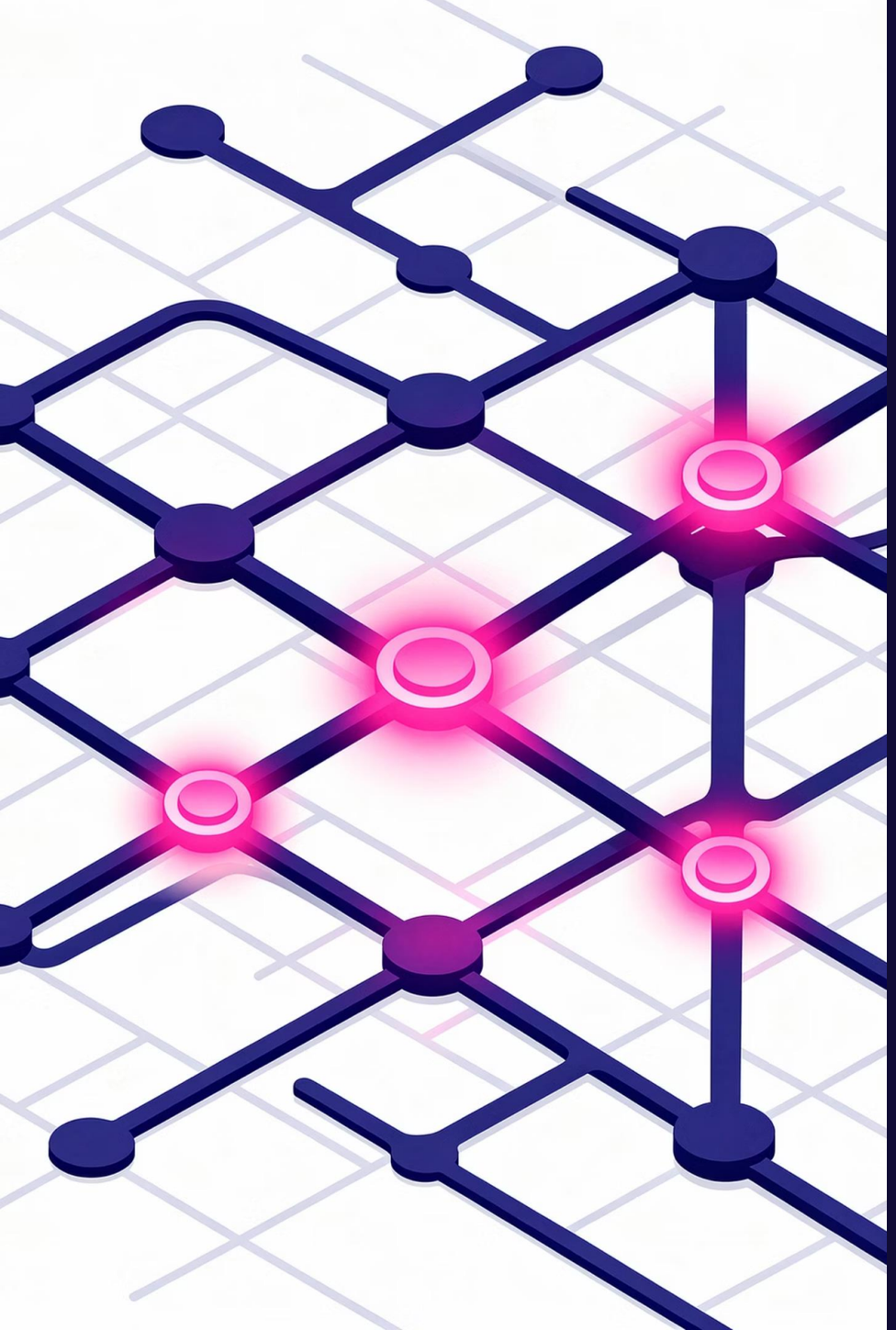
Type

Computer worm



Floppy disk containing the Morris Worm source code, preserved at the **Computer History Museum**.

A portable C "grappling hook" component downloaded the main body onto target systems, while loading down other machines as peripheral victims.



Exploited Vulnerabilities

1

Sendmail Debug Hole

Exploited a flaw in the Unix sendmail programme's debug mode

2

Finger Buffer Overflow

Used a buffer overrun in the fingerd network service

3

Rexec/Rsh Trust

Abused transitive trust from passwordless remote execution logins

4

Weak Passwords

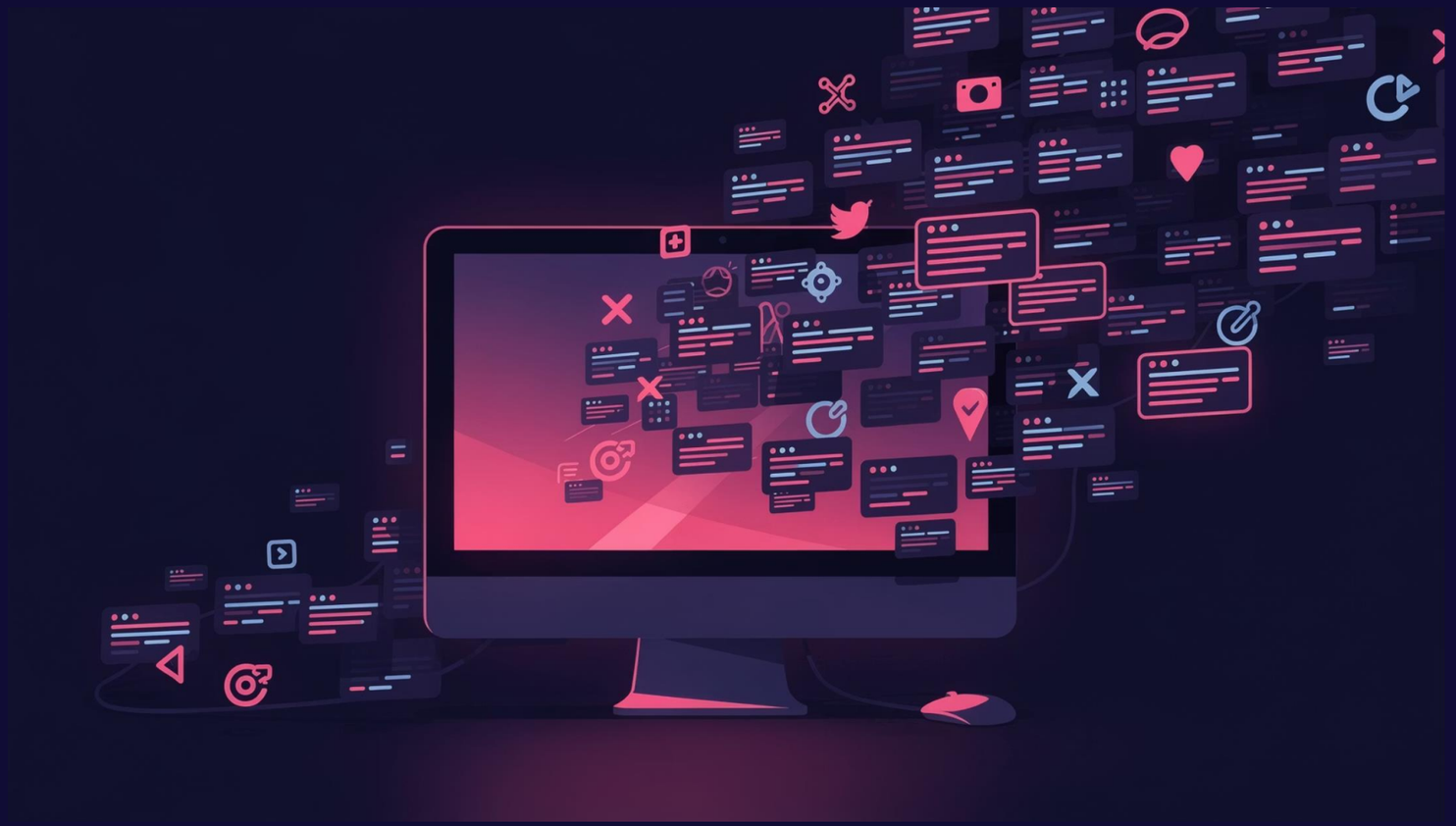
Cracked passwords using dictionary words and simple patterns (e.g. username, usernameusername, emanresu, ...)

From Experiment to Catastrophe

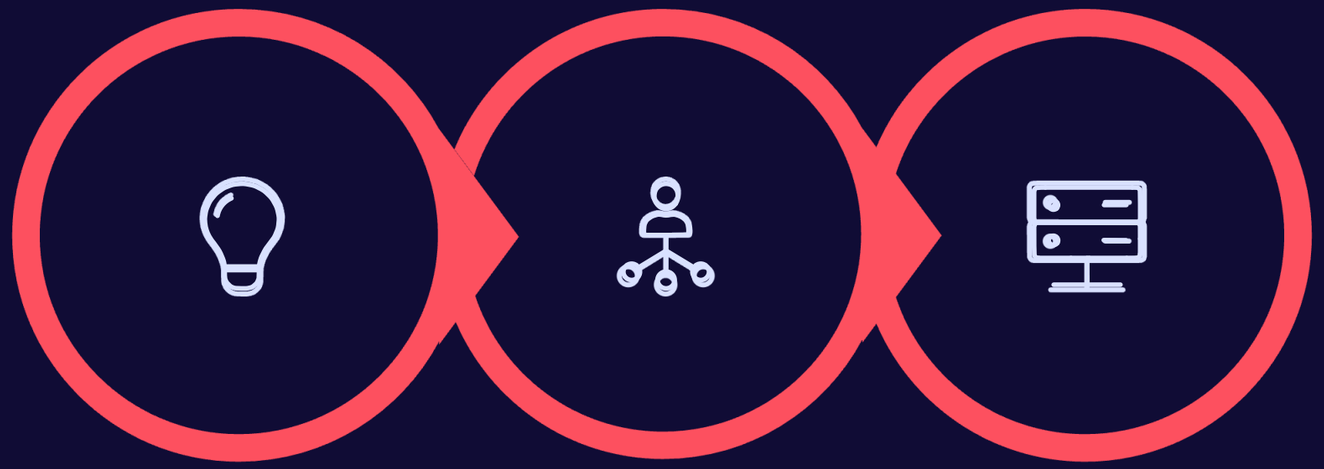
Morris did **not** intend the worm to be destructive — he wanted to highlight network weaknesses. But a critical design decision changed everything.

Fearing administrators would fake "already infected" reports, he programmed the worm to copy itself 14% of the time regardless of infection status.

The result: machines were infected multiple times, each copy slowing them further — effectively creating a fork bomb that crashed computers repeatedly.



Replication Out of Control



Morris: 14% Replication

Worm Spreads Rapidly

Machines Crash (DoS)

> "He should have tried it on a simulator first." — **Michael Rabin**, whose randomisation concept inspired the replication rate

The Damage

~6K

Machines Infected

Major UNIX systems hit (~10% of the internet)

2K

In 15 Hours

Clifford Stoll's direct survey count

$10M

Estimated Cost

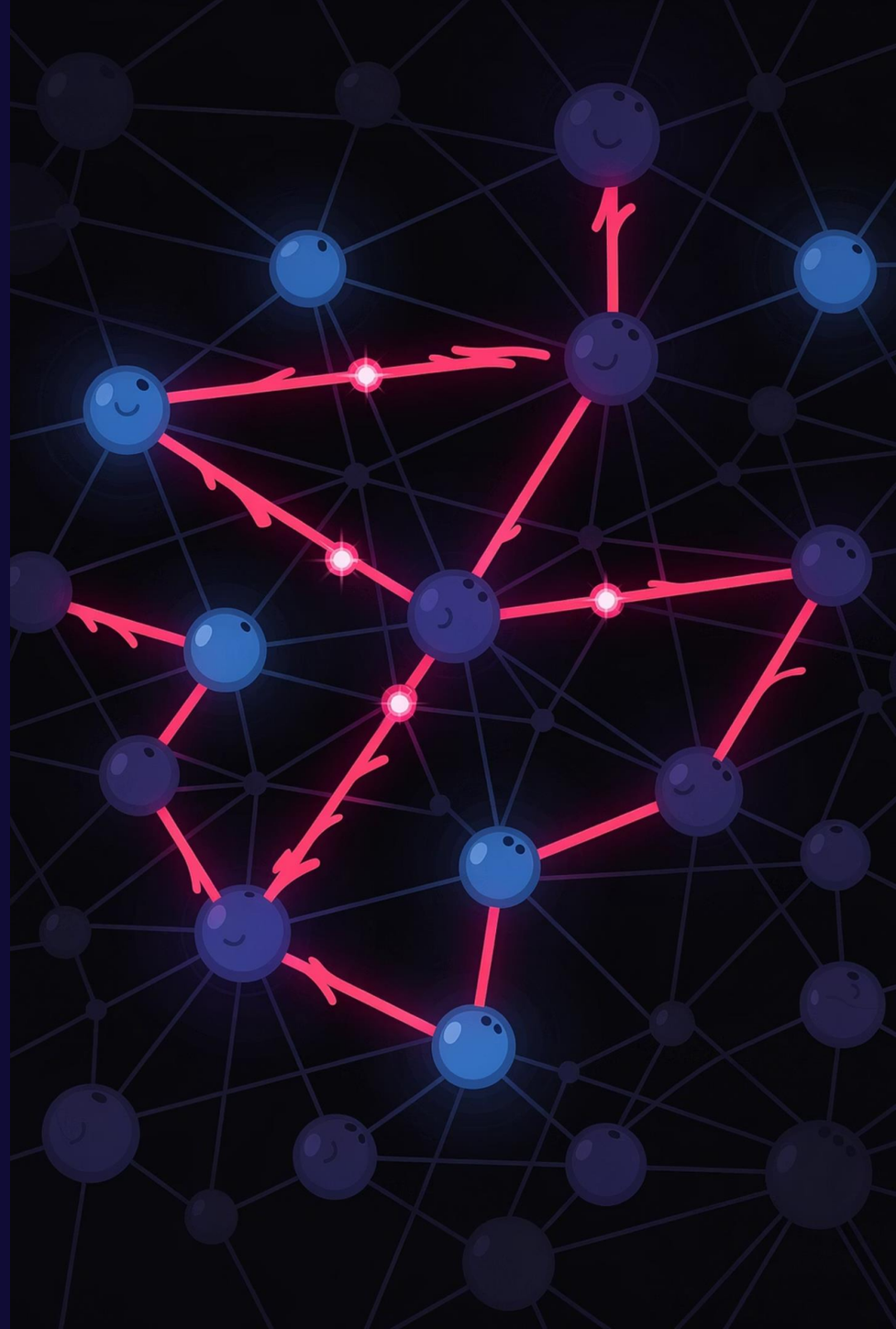
Upper range of total economic impact

$53K

Per Installation

Maximum removal cost per machine

> "I surveyed the network, and found that two thousand computers were infected within fifteen hours. These machines were dead in the water — useless until disinfected." — Clifford Stoll



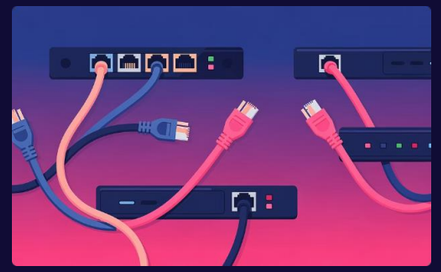
Legacy & Consequences

First Felony Conviction

Morris was convicted under the Computer Fraud and Abuse Act — sentenced to 3 years' probation, 400 hours community service, and fined $10,050.

CERT/CC Founded

DARPA funded the CERT Coordination Centre at Carnegie Mellon University to coordinate responses to network emergencies.



Internet Partitioned

Regional networks disconnected from the NSFNet backbone for days to prevent recontamination while cleaning their systems.

The "Great Worm" in Popular Culture

Named after Tolkien's devastating dragons, the Morris worm left a lasting mark on both security and storytelling.

Hackers (1995)

Features a 1988 viral attack mirroring the Morris worm — infecting 1,000+ computers, causing economic disruption, and resulting in probation.

The Cuckoo's Egg

Clifford Stoll details his efforts battling the worm in the epilogue of his celebrated book.

Halt and Catch Fire

A similarly functioning virus is created to gauge the size of the network in this acclaimed TV series.

Digital: A Love Story

The visual novel portrays the Morris worm as a cover story for a large-scale attack on ARPANET and bulletin board systems.

Lessons That Shaped the Internet

Security Matters

Weak passwords and unpatched software are existential threats

Monoculture Risk

when many systems share the same architecture, software, or dependency, a single vulnerability can propagate rapidly, amplifying impact from isolated failure to large-scale disruption.

Coordinated Response

CERT/CC became the model for incident response worldwide

> "*If all the systems on the ARPANET ran Berkeley Unix, the virus would have disabled all fifty thousand of them.*" — Clifford Stoll, 1989